

MSDM 5056 Network Modeling
Assignment 2 (due 15th October, 2025)

Submit your assignment solution on Canvas. You may discuss with others or seek help from your TA, but you should not directly copy from others. Otherwise, it will be considered as plagiarism.

(1) Matrix formalism

Let $\mathbf{A}$ be the adjacency matrix of an undirected network of size N and $\mathbf{1}$ be the column vector whose elements are all 1, i.e.

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$$

Referring to p.30 of Lecture 2, show that

(a) the vector $\mathbf{k}$ whose elements are the degrees k_i of the vertices, i.e.

$$\mathbf{k} = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_N \end{pmatrix}$$

is given by $\mathbf{k} = \mathbf{A}\mathbf{1}$;

(b) the number L of links in the network is given by

$$L = \frac{1}{2} \mathbf{1}^T \mathbf{A} \mathbf{1}$$

(c) the matrix $\mathbf{C}$ whose element C_{ij} is equal to the number of common neighbors of nodes i and j can be written as

$$\mathbf{C} = \mathbf{A}^2$$

Solution:

(a) The degree of a node in a simple network is given by

$$k_i = \sum_{j=1}^N A_{ij} = \sum_{j=1}^N A_{ij} 1_j$$

It now follows that in vector form, $\mathbf{k} = \mathbf{A}\mathbf{1}$.

(b) The total number of links in the network is given by

$$L = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N A_{ij} = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N 1_i A_{ij} 1_j$$

It follows that L can then be written as $L = \frac{1}{2} \mathbf{1}^T \mathbf{A} \mathbf{1}$.

- (c) A node r is a common neighbor of both nodes i and j if and only if $A_{ir}A_{rj} = 1$. The total number of common neighbors of nodes i and j is $C_{ij} = \sum_{r=1}^N A_{ir}A_{rj}$. One can then write it in matrix form, $\mathbf{C} = \mathbf{A}^2$.

(2) Structure and centrality measures for a directed network

The adjacency matrix $\mathbf{A}$ of a network with 4 nodes is given by

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

- (a) Write the in-degree sequence $\{k_1^{in}, k_2^{in}, k_3^{in}, k_4^{in}\}$ and the out-degree sequence $\{k_1^{out}, k_2^{out}, k_3^{out}, k_4^{out}\}$.
(b) Write the in-degree distribution $P^{in}(k)$, the out-degree distribution $P^{out}(k)$, and the undirected degree distribution $P(k)$ of the network.
(c) The Katz centrality vector $\mathbf{x}$ has elements x_i indicating the Katz centrality of node $i = 1, 2, \dots, N$. Calculate $\mathbf{x}$ using the following definition

$$\mathbf{x} = \beta(\mathbf{I} - \alpha\mathbf{A})^{-1}\mathbf{1} = \beta \sum_{n=0}^{\infty} \alpha^n \mathbf{A}^n \mathbf{1}$$

where $\alpha > 0$, $\beta > 0$, $\mathbf{1}$ is the column vector whose elements are all 1, and $\mathbf{I}$ denotes the $N \times N$ identity matrix.

Solution:

- (a) The in-degree sequence $\{k_1^{in}, k_2^{in}, k_3^{in}, k_4^{in}\} = \{0, 2, 1, 1\}$. The out-degree sequence $\{k_1^{out}, k_2^{out}, k_3^{out}, k_4^{out}\} = \{2, 0, 1, 1\}$
(b) The in-degree distribution $P^{in}(k)$ is $P^{in}(0) = \frac{1}{4}$, $P^{in}(1) = \frac{1}{2}$, $P^{in}(2) = \frac{1}{4}$, and $P^{in}(k) = 0$ for $k > 2$. The out-degree distribution $P^{out}(k)$ is $P^{out}(0) = \frac{1}{4}$, $P^{out}(1) = \frac{1}{2}$, $P^{out}(2) = \frac{1}{4}$, and $P^{out}(k) = 0$ for $k > 2$. The undirected degree distribution is $P(2) = 1$ and $P(k) = 0$ for $k \neq 2$.
(c) To obtain the Katz centrality, we need to compute $\mathbf{A}^n$. First, $\mathbf{A}^0 = \mathbf{I}$.

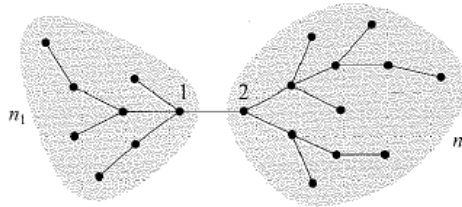
$$\mathbf{A}^2 = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Therefore, $\mathbf{A}^n = 0$ for all $n \geq 3$. Putting everything together, we have

$$\begin{aligned} \mathbf{x} &= \beta \left[\underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_{\mathbf{I}} \mathbf{1} + \alpha \underbrace{\begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}}_{\mathbf{A}} \mathbf{1} + \alpha^2 \underbrace{\begin{pmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{\mathbf{A}^2} \mathbf{1} \right] \\ &= \beta \begin{pmatrix} 1 + 2\alpha + 2\alpha^2 \\ 1 \\ 1 + \alpha \\ 1 + \alpha \end{pmatrix} \end{aligned}$$

(3) Closeness centrality

Consider an undirected tree of n vertices. A particular edge in the tree joins vertices 1 and 2 and divides the tree into two disjoint regions of n_1 and n_2 vertices as sketched below.



Show that the closeness centralities c_1 and c_2 of the two vertices are related by

$$\frac{1}{c_1} + \frac{n_1}{n} = \frac{1}{c_2} + \frac{n_2}{n}$$

(Hint: Begin with the definition of closeness centrality on p.32 of Lecture 3)

Solution:

Let R_1 and R_2 be the region of vertex 1 and that of vertex 2. We see that for any vertex j ,

$$d_{1j} - d_{2j} = \begin{cases} -1 & (j \in R_1) \\ 1 & (j \in R_2) \end{cases}$$

Using the definition of closeness centrality, $\frac{1}{c_i} = \frac{1}{n} \sum_{j=1}^n d_{ij}$ or $\frac{n}{c_i} = \sum_{j=1}^n d_{ij}$, we get

$$\frac{n}{c_1} - \frac{n}{c_2} = \sum_{j=1}^n d_{1j} - \sum_{j=1}^n d_{2j} = \sum_{j \in R_2} (d_{1j} - d_{2j}) + \sum_{j \in R_1} (d_{1j} - d_{2j}) = n_2 - n_1$$

Rearranging, we obtain

$$\frac{1}{c_1} + \frac{n_1}{n} = \frac{1}{c_2} + \frac{n_2}{n}$$

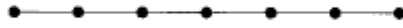
(4) Betweenness centrality

Consider an undirected (connected) tree of n vertices. Suppose that a particular vertex in the tree has degree k so that its removal would divide the tree into k disjoint regions, then suppose that the sizes of those regions are $n_1, n_2, \dots, n_k$.

- (a) Show that the unnormalized betweenness centrality x of the vertex, which counts the paths that connect any two other vertices through it (*refer to p.21 of Lecture 3*), is given by

$$x = \frac{1}{2} \left[(n-1)^2 - \sum_{m=1}^k n_m^2 \right]$$

- (b) Hence calculate the betweenness of the i^{th} vertex from the end of a “line graph” of n vertices arranged in a row like the following example, which has $n = 7$:



Solution:

- (a) The total number of paths that connect any two vertices from different regions through the particular vertex would be $\sum_{i \neq j}^k n_i n_j$. Since there is only one geodesic path for each pair of nodes for an undirected tree, the expression is also equal to the betweenness of the vertex. Thus

$$x = \sum_{i \neq j}^k n_i n_j$$

Adding and subtracting $\frac{1}{2} \sum_{m=1}^k n_m^2$ in the above equation, we have

$$x = \frac{1}{2} (n_1 + n_2 + \dots + n_k)^2 - \sum_{m=1}^k \left(\frac{n_m^2}{2} \right) = \frac{1}{2} \left[(n-1)^2 - \sum_{m=1}^k n_m^2 \right]$$

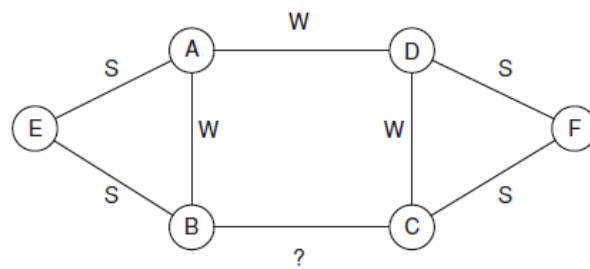
- (b) The two regions on the two sides of the i^{th} vertex have $(i-1)$ and $(n-i)$ vertices respectively. Therefore,

$$x = \frac{1}{2}[(n-1)^2 - (i-1)^2 - (n-i)^2] = (i-1)(n-i)$$

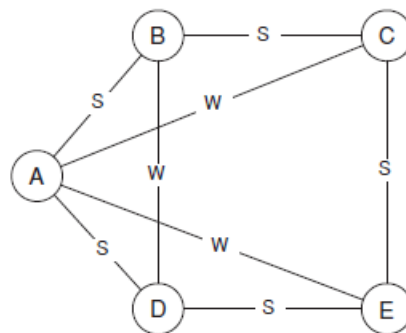
(5) Strong and weak ties

Edges in the following graphs are labeled as either a strong tie (S) or a weak tie (W).

- (a) How may the edge between B and C be labeled if all nodes satisfy the Strong Triadic Closure property? Briefly explain.



- (b) Which two nodes violate the Strong Triadic Closure property? Briefly explain.



Solution:

- (a) The edge may be a weak tie, or it may not be an edge at all. If the edge were a strong tie, there should be an edge between nodes C and E and another edge between B and F.
- (b) Since the edges BC and CE are strong ties, edge BE should exist, and its absence means node C violates the Strong Triadic Closure property. Likewise, since the edges CE and ED are strong ties, edge CD should exist, and its absence means node E violates the property.